

An Unbounded Oriented One-Way LOCC Gap by Subsystem Swap

Candidate full answer to the first question in Section 7.3 of arXiv:1406.1959

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Status. Candidate full result, likely valid, pending human review. For the oriented one-way class $\text{LOCC}^{A \rightarrow B}$ displayed in the source, there are bipartite state pairs whose LOCC distinguishability norm equals 2 while their $\text{LOCC}^{A \rightarrow B}$ norm is $O(d^{-1/2})$. Thus no dimension-free comparison $\|\cdot\|_{\text{LOCC}^{A \rightarrow B}} \geq c \|\cdot\|_{\text{LOCC}}$ is possible.

1 Source question and convention

Aubrun and Lancien define the oriented one-way class $\text{LOCC}^{A \rightarrow B}$ by POVMs of the form

$$(M_i \otimes N_{i,j})_{i,j}, \quad \sum_i M_i = I_A, \quad \sum_j N_{i,j} = I_B,$$

so Alice measures first and Bob's measurement may depend on Alice's outcome [1, Section 2.3]. Section 7.3 asks whether the gap between this class and unrestricted finite-round LOCC can be unbounded, or whether an absolute $c > 0$ satisfies

$$\|\cdot\|_{\text{LOCC}^{A \rightarrow B}} \geq c \|\cdot\|_{\text{LOCC}} \tag{1}$$

in every local dimension.

is between $\exp(\Omega(d^*))$ and $\exp(O(d^*))$.

Let us make another comment on that topic. Theorem 2 tells us, amongst other, that the class of PPT POVMs is, in some sense, a quite good approximation of the class of all POVMs. One may therefore wonder if there would be a way, when trying to approximate the class of all POVMs by a finite sub-family, to impose that all POVMs in it are PPT. However, since the approximation we are looking for is one in terms of distinguishability norms (i.e. one that is valid for *any* pair of states to be discriminated), this possibility is ruled out by the fact that the gap between $\|\cdot\|_{\text{PPT}}$ and $\|\cdot\|_{\text{ALL}}$ is unbounded (i.e. that there exist pair of states, such as e.g. the Werner states, which are poorly distinguished by any PPT POVM).

7.2. What is the typical performance of the class $\underline{\text{LO}}$?

While Theorem 3 shows that the gap between the classes $\underline{\text{LO}}$ and $\underline{\text{LOCC}}$ may be unbounded, we do not know if this situation is typical or not. Asking whether norms are comparable in a typical direction is more or less equivalent to asking whether the ratio $\text{vrad}(K_{\underline{\text{LOCC}}})/\text{vrad}(K_{\underline{\text{LO}}})$ is bounded as the dimension increases.

7.3. Can the gap between $\underline{\text{LOCC}}^{\rightarrow}/\underline{\text{LOCC}}/\underline{\text{SEP}}$ be unbounded?

Or conversely, does there exist an absolute constant c such that the norm inequalities $\|\cdot\|_{\underline{\text{LOCC}}^{\rightarrow}} \geq c \|\cdot\|_{\underline{\text{LOCC}}}$ and/or $\|\cdot\|_{\underline{\text{LOCC}}} \geq c \|\cdot\|_{\underline{\text{SEP}}}$ hold for any dimension?

7.4. Locally restricted measurements on a multipartite quantum system

There are at least two ways for a multipartite system such as $(\mathbb{C}^d)^{\otimes k}$ to be of high

Figure 1: The printed question in Section 7.3 of the source, arXiv PDF page 20.

The direction matters. We write $\text{LOCC}^{B \rightarrow A}$ for the analogous class with Bob measuring first. The result below answers the question for the oriented class exactly as displayed. It does not address a different convention in which “one-way LOCC” means the union of the two orientations.

For a POVM $\mathbf{M} = (M_a)_a$ and a Hermitian operator Δ , recall

$$\|\Delta\|_{\mathbf{M}} = \sum_a |\text{Tr}(\Delta M_a)|,$$

and the norm of a measurement class is the supremum over its POVMs.

2 Reverse communication collapses on classical-quantum inputs

Lemma 1 (Classical-side collapse). *Let*

$$\Delta = \sum_{i=1}^d |i\rangle\langle i| \otimes \Delta_i \in \mathcal{H}(\mathbb{C}^d \otimes \mathbb{C}^d), \quad (2)$$

where every Δ_i is Hermitian. Then

$$\|\Delta\|_{\text{LOCC}^{B \rightarrow A}} = \|\Delta\|_{\text{LO}} = \sup_{\mathbf{N}=(N_j)_j} \sum_{i=1}^d \|\Delta_i\|_{\mathbf{N}}. \quad (3)$$

Proof. The last expression is attained within LO: Alice measures in the basis $(|i\rangle\langle i|)_i$ and Bob uses $\mathbf{N}$. Conversely, consider an arbitrary $B \rightarrow A$ one-way POVM. Its effects have the form

$$M_{j,k} \otimes N_j,$$

where $\mathbf{N} = (N_j)_j$ is Bob’s first measurement and, for every j , $(M_{j,k})_k$ is Alice’s conditional POVM. Put $m_{j,k,i} = \langle i|M_{j,k}|i\rangle$. These numbers are nonnegative and $\sum_k m_{j,k,i} = 1$. Hence

$$\begin{aligned} \sum_{j,k} |\text{Tr}(\Delta(M_{j,k} \otimes N_j))| &= \sum_{j,k} \left| \sum_i m_{j,k,i} \text{Tr}(\Delta_i N_j) \right| \\ &\leq \sum_{j,k,i} m_{j,k,i} |\text{Tr}(\Delta_i N_j)| \\ &= \sum_{i,j} |\text{Tr}(\Delta_i N_j)| = \sum_i \|\Delta_i\|_{\mathbf{N}}. \end{aligned}$$

Taking the supremum gives $\|\Delta\|_{\text{LOCC}^{B \rightarrow A}} \leq \|\Delta\|_{\text{LO}}$, while the reverse inequality follows from $\text{LO} \subset \text{LOCC}^{B \rightarrow A}$. $\square$

The lemma says that communication arriving at a party whose register is already classical cannot improve the norm: any conditional processing of that classical register is dominated by reading it outright at the start.

3 The swapped data-locking pair

The source proves the following theorem. There is a universal constant C such that, for every local dimension d , one can find states ρ_d, σ_d on $\mathbb{C}^d \otimes \mathbb{C}^d$ with

$$\|\rho_d - \sigma_d\|_{\text{LOCC}^{A \rightarrow B}} = 2, \quad \|\rho_d - \sigma_d\|_{\text{LO}} \leq \frac{C}{\sqrt{d}}. \quad (4)$$

Moreover, their construction is classical on Alice's side:

$$\rho_d = \frac{1}{d} \sum_{i=1}^d |i\rangle\langle i| \otimes \rho_i, \quad \sigma_d = \frac{1}{d} \sum_{i=1}^d |i\rangle\langle i| \otimes \sigma_i, \quad (5)$$

where ρ_i and σ_i have orthogonal supports [1, Theorem 3].

give examples of states which are perfectly distinguishable under local measurements and one-way classical communication but very poorly distinguishable under local measurements with no communication between the parties.

Theorem 3 *There is a universal constant C such that the following holds: for any dimension d , there exists states ρ and σ on $\mathbb{C}^d \otimes \mathbb{C}^d$ such that*

$$\|\rho - \sigma\|_{\text{LOCC}} = 2,$$

and

$$\|\rho - \sigma\|_{\text{LO}} \leq \frac{C}{\sqrt{d}}. \quad (3)$$

These states are constructed as follows: assuming without loss of generality that d is even, let E be a fixed $d/2$ -dimensional subspace of $\mathbb{C}^d$, let $U_1, \dots, U_d$ be random independent Haar-distributed unitaries on $\mathbb{C}^d$, and define the random states $\rho_i = U_i \frac{P_E}{d/2} U_i^\dagger$ and $\sigma_i = U_i \frac{P_{E^\perp}}{d/2} U_i^\dagger$, $1 \leq i \leq d$, on $\mathbb{C}^d$ (where P_E and $P_{E^\perp}$ denote the orthogonal projections onto E and $E^\perp$ respectively). Then, denoting by $\{|1\rangle, \dots, |d\rangle\}$ an orthonormal basis of $\mathbb{C}^d$, define

$$\rho = \frac{1}{d} \sum_{i=1}^d |i\rangle\langle i| \otimes \rho_i \quad \text{and} \quad \sigma = \frac{1}{d} \sum_{i=1}^d |i\rangle\langle i| \otimes \sigma_i.$$

The pair (ρ, σ) satisfies 3 with high probability.

Theorem 3 is proved in Section 4. It is built on the idea that, typically, a single POVM cannot succeed simultaneously in several “sufficiently different” discrimination tasks.

Figure 2: Theorem 3 and the classical-quantum construction in the source, arXiv PDF page 7.

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Let $S(x \otimes y) = y \otimes x$ be the tensor-factor swap and define

$$\tilde{\rho}_d = S \rho_d S^*, \quad \tilde{\sigma}_d = S \sigma_d S^*.$$

Theorem 1 (Unbounded oriented one-way gap). *There is a universal constant C such that*

$$\|\tilde{\rho}_d - \tilde{\sigma}_d\|_{\text{LOCC}} = 2, \quad \|\tilde{\rho}_d - \tilde{\sigma}_d\|_{\text{LOCC}^{A \rightarrow B}} \leq \frac{C}{\sqrt{d}}. \quad (6)$$

Consequently

$$\sup_{0 \neq \Delta \in \mathcal{H}(\mathbb{C}^d \otimes \mathbb{C}^d)} \frac{\|\Delta\|_{\text{LOCC}}}{\|\Delta\|_{\text{LOCC}^{A \rightarrow B}}} \geq \frac{2}{C} \sqrt{d},$$

and the dimension-free inequality (1) is false.

Proof. Swapping the factors converts an $A \rightarrow B$ protocol into a $B \rightarrow A$ protocol and conversely. Therefore Lemma 1, applied to $\rho_d - \sigma_d$ in (5), and (4) give

$$\|\tilde{\rho}_d - \tilde{\sigma}_d\|_{\text{LOCC}^{A \rightarrow B}} = \|\rho_d - \sigma_d\|_{\text{LOCC}^{B \rightarrow A}} = \|\rho_d - \sigma_d\|_{\text{LO}} \leq \frac{C}{\sqrt{d}}.$$

Unrestricted LOCC is invariant under exchanging the parties. The $A \rightarrow B$ protocol in (4) distinguishes ρ_d and σ_d perfectly, so after the swap the corresponding $B \rightarrow A$ protocol is an unrestricted LOCC protocol that distinguishes $\tilde{\rho}_d$ and $\tilde{\sigma}_d$ perfectly. Its norm is therefore 2, the maximum possible for a difference of states. $\square$

4 Scope, verification, and literature boundary

The proof uses no new probabilistic estimate: all asymptotics come from the source’s Theorem 3. The new step is the exact collapse lemma, followed by the tensor-factor swap. The companion script `code/verify_collapse.py` tests the triangle inequality behind the lemma on randomly generated classical-quantum operators and reverse one-way POVMs; the proof itself is the displayed scalar calculation.

Bounded exact-question, exact-title, keyword, and citation searches through 26 August 2026 found work separating one-way and two-way protocols in other hypothesis-testing regimes, but no later primary source recording this $\Omega(\sqrt{d})$ norm separation or observing the subsystem-swap consequence of Theorem 3. The result should be treated as a candidate full answer under the source’s displayed oriented convention, pending expert review.

Remark 1 (Convention boundary). *If $\text{LOCC}^{1\text{-way}}$ is instead defined as $\text{LOCC}^{A \rightarrow B} \cup \text{LOCC}^{B \rightarrow A}$, the swapped pair is perfectly distinguished by its $B \rightarrow A$ member and gives no separation. The stronger union-of-directions question remains open here.*

5 Human review recommendation

Check three points: the source’s arrow is oriented exactly as in its displayed definition; the absolute-value triangle inequality in Lemma 1 permits no adaptive advantage on the classical register; and tensor swap preserves unrestricted LOCC while reversing the one-way orientation. These checks make (6) immediate from the source’s Theorem 3.

References

- [1] G. Aubrun and C. Lancien, *Locally restricted measurements on a multipartite quantum system: data hiding is generic*, Quantum Information and Computation **15** (2015), 513–540; arXiv:1406.1959.